

Course Project Proposal: Group 215

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Problem Statement

This project aims to understand the drivers and trajectories of recovery time for patients with the same medical diagnosis using the MIMIC-IV dataset. Recovery time is defined as the length of stay (LOS) in the hospital or ICU, or the time until clinical stability is achieved. Accurate predictions of recovery time can help hospitals optimize patient care plans, improve the allocation of resources, and enhance discharge planning processes. Additionally, with a better understanding of the factors that most affect recovery time, medical experts can provide preventative guidance to at-risk individuals, and doctors and hospitals can target these factors to potentially reduce LOS. Controlling for the same diagnosis (e.g., pneumonia, heart failure) ensures that the model comparisons are valid by minimizing the impact of confounding factors caused by different illnesses.

Methodology

Data Collection and Preprocessing

The data required for this study will be extracted from the MIMIC-IV dataset. Per online documentation:

*MIMIC (Medical Information Mart for Intensive Care) is a large, freely-available database comprising deidentified health-related data from patients who were admitted to the critical care units of the Beth Israel Deaconess Medical Center.*¹

The dataset contains de-identified health records of patients, including diagnoses, time-series vitals, medications, and demographic details.² The following preprocessing steps will be applied:

- **Filtering Patients:** Select and group patients with the same primary diagnosis using ICD codes from the diagnoses_icd table.
- **Calculating Recovery Time:** Compute LOS or time-to-stability from the admissions and icustays tables.
- **Handling Missing Data:** Impute missing values using forward-fill for vitals and median imputation for lab results.

¹ <https://mimic.mit.edu/docs/iv/modules/>

² <https://mimic.mit.edu/docs/iv/about/concepts/>

- **Feature Engineering:** Create time-based features (e.g., 24-hour averages of vitals) and binary indicators for comorbidities.
- **Data Normalization:** Standardize continuous variables to ensure stable model training and improve convergence.

Model Selection

Several models will be considered for predicting recovery time, with each offering unique advantages. The choice of the final model will depend on performance during initial testing and model interpretability.

- **Linear Regression / Lasso Regression:** To explore linear relationships between features and recovery time.
- **Random Forest / XGBoost:** Capture non-linear patterns and interactions between features.
- **LSTM / GRU Models:** Handle sequential time-series data (e.g., vitals) and capture temporal dependencies.
- **Time Series Decomposition/Forecasting (Facebook Prophet):** Decompose time series data by separating trend, seasonality, etc. Use resulting data for forecasting/predictions.
- **Survival Analysis (Cox Proportional Hazards Model):** Suitable for time-to-event predictions and handling censored data (e.g., patients still in care).
- **Bayesian Analysis:** Estimate the posterior distribution of recovery timelines.

The model selection process will involve performance comparisons using cross-validation. In cases where interpretability is critical for clinical use, simpler models such as linear regression may be preferred. For scenarios involving large time-series data, LSTMs or GRUs may provide better performance by capturing temporal patterns.

Evaluation Strategies

The performance of the selected models will be evaluated using the following metrics and strategies:

- **Mean Absolute Error (MAE):** Measures the average magnitude of prediction errors without penalizing large deviations.
- **Root Mean Square Error (RMSE):** Gives more weight to larger errors, useful when large deviations from actual values are important.
- **C-Index (for Survival Models):** Evaluates how well the model ranks patients by their recovery times.

- **Cross-Validation:** Ensures robustness by splitting the dataset into multiple train-test subsets to avoid overfitting.
- **Propensity Score Matching:** Controls for confounding variables by matching patients with similar demographic and clinical characteristics.
- **Feature Importance Analysis:** Identifies the most influential predictors of recovery time, especially with tree-based models like Random Forest or XGBoost.

Conclusion

This study aims to predict recovery time for patients with specific diagnoses using the MIMIC-IV dataset. The project will leverage both time-series data and patient demographic information to build accurate models. Model selection will focus on balancing predictive accuracy with interpretability, ensuring that the findings are clinically meaningful. Robust evaluation strategies will be used to validate the models, including cross-validation and feature importance analysis. This research will contribute to personalized healthcare planning, better resource allocation, and improved patient outcomes.